Moving Target Indicator Using Neural Networks

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Abstract

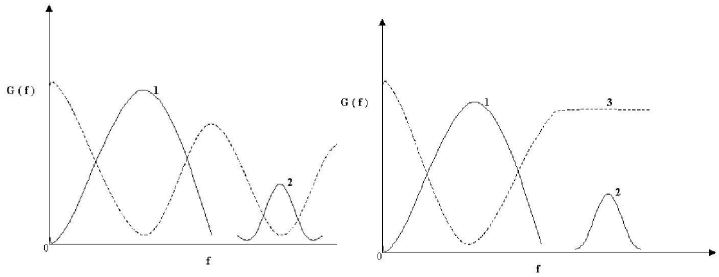
In this paper, we propose a mixed scheme using radial basis function (RBF) neural network for moving target indicator (MTI). Our computer simulated results show that, this mixed scheme using RBF neural network approach is suitable for real-time detection of the moving target indicator. We found that RBF approach is able to detect signal even if signal-to-noise-ratio (SNR) is less than -100 dB while backpropagation (BP) and spectral estimation (SE) approaches fails at -60 dB and -25 dB respectively. We observed that, the performance of detection depends on many factors, such as the bandwidths and the central frequencies of the noise spectrum, and the signal spectrum. This scheme shows improved detection performance compared to backpropagation (BP) and spectral estimation (SE) methods for MTI.

1 Introduction

The use of neural networks finds wide application in signal processing field, such as filtering, parameter estimation, system identification, parameter recognition, signal detection, signal reconstruction, signal compression, moving target indicator, etc. Neural networks has emerged as a strong contender in various application due to unique feature to learn from and adapt to the environment. This lead to use of it into radar signal processing where adaptability plays an important role. It is desirable that the transfer function of the filter traces the power spectrum of the noise as closely as possible so as to reject the noise as much as possible. The filter will minimize not only the power but also the signal power and this is referred to as the signal-cancellation problem. In [1] and [2] signal detection involves inferring from (imperfect) observational data whether or not a target signal is present. In [3] Wilson, Umesh and Tufts employed multistage neural network for transient detection and feature extraction. Liou and Azimi-Sadjadi [4] has shown that dim targets can be detected using high order correlation method. For the purpose of improving the detection performance, researchers utilize some preprocessing or post-processing, such as pulse-compression [5], [6] and constant-false-alarm-rate processing involving the likelihood-ratio detector. The preprocessing and the post-pr ocessing are usually computationally complex or highly nonlinear, and neural networks are also very powerful here. In [7] Roth used neural networks for extraction of weak targets in high clutter environments. In [8], [9] an effective detection method based on the spectral estimation is proposed. Here, we propose a mixed scheme using RBF neural network for moving-target indicator and detection performance is compared to BP and SE method. The computer simulations demonstrate the suitability of mixed scheme using RBF neural network for moving-target indicator.

2 Problem Formulation and Neural Network for MTI

In moving target indicator, we usually first use the difference of the power spectra of the moving target and the noise to design a noise rejection filter and then take the output of the noise rejection filter as the observation (input) of the likelihood ratio detector to make a detection. Since the output of the noise rejection filter has a much higher signal-to-noise ratio than the input, the detection performance with noise-rejection filter is much better than without the filter. The probability density function of the output of the noise rejection filter is Gaussian. As a result, the corresponding likelihood ratio detector approximate a matched filter detector. In signal-cancellation prob lem the filter will minimize not only the noise power but also the signal power. This is referred to as the signal-cancell ation problem. Figure 1-(a) and 1-(b) show the spectral relationship of the filter with the signal cancellation and without signal cancellation. An effective detection method based on the spectral estimation is proposed in [8], however, this method based on spectral estimation involves very extensive computation. In this paper, our main focus is to investigate the movin g target indicator method to overcome the signal-cancellation problem and to minimize the required computation. Neural networ k for spectral estimation offer new opportunities to implement this method.



(a) Signal-cancellation. (b) Without signal-cancellation.

Figure 1: Problem with and without signal-cancellation.

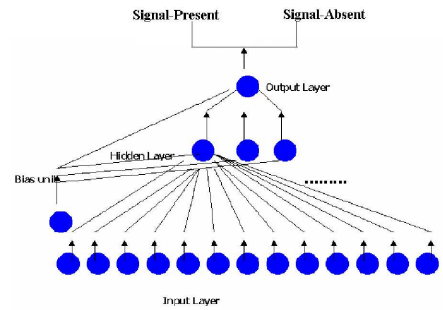


Figure 2: BP Neural Network

2.1 Spectral Estimation Approach

To overcome the signal-cancellation problem, an effective detection method based on the spectral estimation is used. This method includes three steps:

- Estimation of the spectrum (or parameters) of the observation samples.
- Calculation of the frequency values of the peaks of the estimated spectrum.
- Determination of the whether the target signal is present or not according to these frequency values.

However, because this method based on spectral estimation involves very extensive computation, it was impractical in the past. Neural networks for spectral estimation offer new opportunities to implement this method. We will show how to use neural networks to implement it. Tracking of objects always takes place at unknown regions and hence the learning ability of neural networks can be used to maximum capacity. Neural networks can be trained with a particular training set before being exposed and changed of training sets can be done to make it suitable for different environments.

2.2 Backpropagation Algorithm For MTI

We use a multi-layer feedforward neural network with a backpropagation algorithm for moving target indicator. The idea is t o train the network, so that it will calculate the real-time filter coefficients. The structure of the BP neural network fo r order $P = 6$, is shown in Figure 2. Each of the networks used has n input units. In our experiments $n = 13$. Three hidd en units, and one output unit. The output unit is classifier. The learning rule used is backpropagation with gradiant decen t with biases training. The sigmoid function is used as the activation function of the neurons in the hidden and output lay ers and direct connection is used in the input layer. A bias unit, which always has the value 1, is connected to all neuron s through weights in the hidden and the output layers to achieve a better calculation of the coefficients of the noise reje ction filter.

2.3 Proposed RBF Neural Network Approach For MTI

The structure of propose RBF neural network for MTI is shown in Figure 3. The input layer has $N = 13$ neurons with a linear function. One hidden layer of K neurons with Gaussian transfer functions. The output layer has only one neuron whose input-output relationship approximates the two possible states. The two bias nodes are included as part of the network. A real-valued input vector $\mathbf{x}$ to a neuron of the hidden or output layer produces neural output $G(\mathbf{x})$, where $0 < G(\mathbf{x}) < 1$. The Gaussian function we choose is $G(\mathbf{x}, \mathbf{x}_i) = exp(-1/2\sigma_i^2 \|\mathbf{x} - \mathbf{x_i}\|^2)$. The RBF neural network MTI coefficients statistic $T_{NN}(\mathbf{x})$ may now be expressed as,

$$T_{NN}(\mathbf{x}) = \sum_{i=1}^{K} w_i \varphi_i(\mathbf{x}) + w^{b_2} \tag{1}$$

where $\{\varphi_i(\mathbf{x}), i = 1, 2, 3, ..., K\}$ is a set of basis functions. The $\mathbf{w}$ constitutes a set of connection weights for the output layer. When using RBF the basis is

$$\varphi_i(\mathbf{x}) = G(\|\mathbf{x} - \Sigma \mathbf{t_i}\|^2) + w_i^{b_1}, i = 1, 2, 3, ..., K \tag{2}$$

where $\mathbf{t}_i = [\mathbf{t}_{i1}, \mathbf{t}_{i2}, ..., \mathbf{t}_{iN}]^T$ with $\mathbf{t}_i$ as unknown centers to be determined. Σ is a symmetric positive definite weighting matrix of size $N\mathbf{x}N$, $G(.)$ represents a multivariate Gaussian distribution with mean vector $\mathbf{t}_i$ and covariance matrix Σ. By using above equations we redefine coefficient statistics $T_{NN}(\mathbf{x})$ as

$$T_{NN}(\mathbf{x}) = \sum_{i=1}^{K} w_i G(\mathbf{x}, \mathbf{t}_i) = \sum_{i=1}^{K} w_i G(\|\mathbf{x} - \mathbf{t}_i\|). \tag{3}$$

We determine the set of weights $\mathbf{w} = [w_1, w_2, ..., w_k]^T$ and the set $\mathbf{t}$ of vectors $\mathbf{t}_i$ of centers.

The coefficient of the noise-rejection filter are exactly the same as those used in the spectral estimation. That is the output $u(n)$ of the noise rejection filter can be written as,

$$u(n) = x(n) - \sum_{k=1}^{p} a_k x(n-1). \tag{4}$$

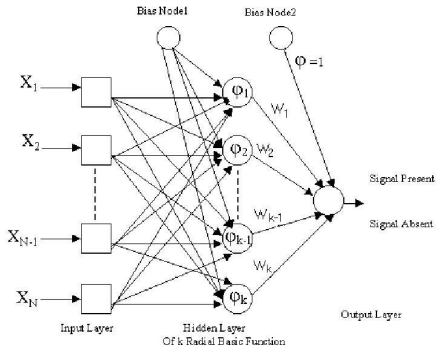


Figure 3: A RBF Neural Network

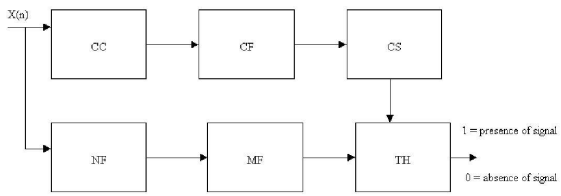


Figure 4: A mixed scheme using RBF neural network for moving target indicator.

This means that we can use the neural network for least square algorithm for the real time coefficients of the noise rejection filter. Based on this argument, we can devise a mixed scheme for moving target indicator, as shown in Figure 4. In Figure 4, CC: computation of the coefficients by neural networks; CF: computation of frequency values of the roots; CS: computation of the sufficient statistic; NF: the noise-rejection filter; MF: the matched filter; TH: thresholding. The high-speed computational capability of the neural network plays an important role in moving target indicator. If the received signal consists of $x(0), x(1), ..., x(N-1)$ samples. Using the received signal we generate the data matrix $\mathbf{X}$ and data vector $\mathbf{X_1}$. where

$$\mathbf{X} = \begin{pmatrix} x(0) & 0 & \cdots & 0 \\ x(1) & x(0) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ x(P-1) & x(P-2) & \cdots & x(0) \\ \vdots & \vdots & \ddots & \vdots \\ x(N-1) & x(N-2) & \cdots & x(N-P) \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \vdots & x(0) & x(1) \\ 0 & \cdots & 0 & x(0) \end{pmatrix}$$

$$\mathbf{X_1} = -\begin{pmatrix} x(1) \\ x(2) \\ \vdots \\ x(N-1) \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

and

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{pmatrix}$$

$\mathbf{a}$ is the coefficient vector. Using $\mathbf{X}$, $\mathbf{X}_1$ and $\mathbf{a}$ we write,

$$\mathbf{X}^T\mathbf{X}\mathbf{a} = \mathbf{X}^T\mathbf{X}_1 \tag{5}$$

The solution of equation (5) is

$$\mathbf{a} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}_1 \tag{6}$$

Using coefficients statics we can find filter coefficients a_k (for $k = 1, 2, 3, ..., p$)and from filter coefficients we can calculate the roots of the polynomial

$$1 - \sum_{k=1}^{p} a_k z^{-k} \tag{7}$$

and the frequency values of the peaks corresponding to these roots are

$$f_i = \frac{1}{2\pi T_r} arctan[\frac{I_m(z_i)}{R_e(z_i)}], (i = 1, 2, ..., p) \tag{8}$$

If the signal is absent all roots will be located closely to $z_c = e^{j2\pi f_c T_r}$. If the signal is present and the signal-to-noise ratio is not too small, then there is at least one root that is closer to $z_s = e^{j2\pi f_s T_r}$ than to $z_c = e^{j2\pi f_c T_r}$. From these, we can formulate a sufficient statistics v as

$$v = \max_{i,j} |f_i - f_j|, (i, j = 1, 2, 3, ..., p) \tag{9}$$

$$v > n_f \Rightarrow H_1 \tag{10}$$

$$v \leq n_f \Rightarrow H_0 \tag{11}$$

That is we can declare the si gnal to be present if there are two different roots; otherwise, the hypothesis of the absence of the signal is chosen.

2.4 Training The Network

During each training epoch, both noise-only and signal-plus-noise values are presented to the network. For training we consider 100000 noise-only inputs and 25000 signal-plus-noise only inputs. During noise-only inputs, the network is trained to produce the filter coefficients where booth roots will be located closely to $z_c = e^{j2\pi f_c T_r}$ while, during signal-plus-noise inputs, it is trained to produce the filter coefficients where there is at least one root that is closer to $z_s = e^{j2\pi f_s T_r}$ than to $z_c = e^{j2\pi f_c T_r}$. Once the network is trained now it is employed for filter

coefficient calculations, by feeding sequences which contains noise-only or signal-plus-noise. Using these filter coefficients we calculate the frequency values of peaks corresponding to these roots. And by using sufficient statistics v we can declare presence of the signal if there are two different roots, otherwise the hypothesis of the absence of the sig nal is chosen.

3 Experimental Results and Performance of Mixed Scheme

To illustrate the potential of this new class of MTI, we now present simulated performance comparison of moving target indicator using SE, BP and RBF neural network, by considering different noise and signal spectrum bandwidth. While calculating real-time coefficients we consider a received signal consisting of P sinusoids (where P = 6) corrupted by white noise. $x(n) = \sum_{i=1}^{P} \alpha_i \cos(\omega_i n) + \epsilon(n) \quad (n = 1, 2, 3, ..., L)$ where $x(n)$ and $\epsilon(n)$ are the measured samples and noise samples respectively. The parameters α_i, ω_i denote the amplitude, frequency of the i'th sinusoid. $\epsilon(n)$ is zero mean and has variance σ^2.

3.1 Simulation Parameter Selection

In simulation ,we selected the order of the AR model $P = 6$. Number of samples in one sinusoid $n = 13$. The propose RBF neural network has $N = 13$ input nodes, $K = 13$ hidden nodes. It is trained by 25000 epochs. Sinusoidal signals corrupted by white Gaussian noise. Test set contains 100000 noise-only vectors and 25000 signal-plus-noise only vectors. The sampling period $T_r = 1$. B_c and B_s as the bandwidths of the noise spectrum and the target signal spectrum respectively. Here we considered Noise bandwidth spectrum as an threshold. The central frequency values of $f_c = 100$ Hz and $f_s = 300$Hz ,where f_c and f_s represent the central frequency values of noise and signal respectively. SNR is the signal-to-noise ratio varies using noise variance. The relationships between the estimated frequency values of the roots and the SNR for different bandwidths of the noise spectrum (B_c) and target signal spectrum (B_s) are plotted.

3.2 Estimated Frequency Roots

Here we present the simulation results of estimated complex-conjugate frequency roots calculated by SE, BP and RBF neural network approach by varying SNR. While estimating these roots we consider the condition $|f_s - f_c| \leq B_c$. It states that any signal whose central frequency value f_s satisfies this condition will be missed. Figure 5 show the estimated complex-conjugate roots for 1000 noise only vectors. Results show that we are able to estimate the frequency roots up to -45 dB, -75dB and -85 dB for SE, BP and RBF approach respectively. Results show that we are able to detect frequency roots better than SE and BP.

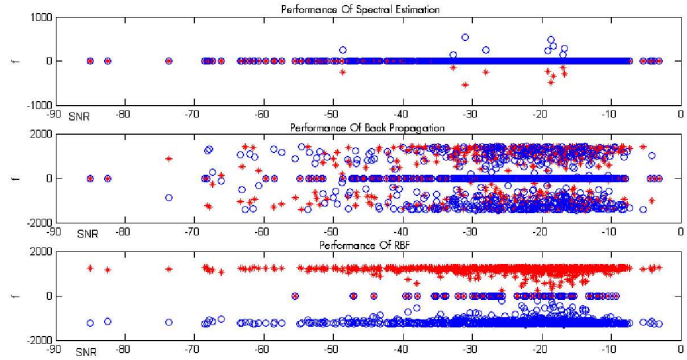


Figure 5: Detection of frequency roots at 1000 noise only vectors by SE, BP and RBF approach.

3.3 Performance of MTI Using SE ,BP and RBF Neural Network

Here we, illustrate performance comparisons of the moving-target indicator using SE, BP and RBF neural network. In these simulations, we considered AR model $P = 6$. Number of samples in one sinusoid $n = 13$. The RBF neural network has $N = 13$ input nodes, $K = 13$ hidden nodes. It is trained by 25000 epochs. Sinusoidal signals corrupted by white Gaussian noise. Test set contains 100000 noise-only vectors and 25000 signal-plus-noise only vectors. The sampling period $T_r = 1$. B_c and B_s as the bandwidths of the noise spectrum and the target signal spectrum respectively. Here we considered Noise bandwidth spectrum as an threshold. The central frequency values of $f_c = 100$ Hz and $f_s = 300$Hz ,where f_c and f_s represent the central frequency values of noise and signal respectively. SNR is the signal-to-noise ratio varies using noise variance. For different bandwidths of the noise spectrum B_c and the target signal spectrum B_s and by varying signal-to-noise ratio we estimated the corresponding frequency values of roots. Figure 6 show the relationship between the estimated frequency values of the roots and the SNR for SE, BP and RBF respectively. Curves 1 and 3 represent the central frequency values of the noise and signal respectively. Curves 2 and 4 represent the estimated frequency values of the two roots. In case, of SE approach we are able to detect targets up to -50 dB and -82 dB SNR respectively while, in case of RBF neural network approach the signal can be found even if SNR = -100 dB as illustrated in Figure 6. Figures 7, 8 and 9 show the comparison of SE, BP and RBF neural network performance for different set of values of bandwidths of the noise spectrum and the target signal spectrum. These facts suggest that we should mix RBF neural network approach so as to deliver better performance for mov ing-target indicator. Table I show the performance comparasion between SE, BP and RBF MTI. These facts suggest that we should mix RBF neural network approach as to deliver better performance for moving-target indicator.

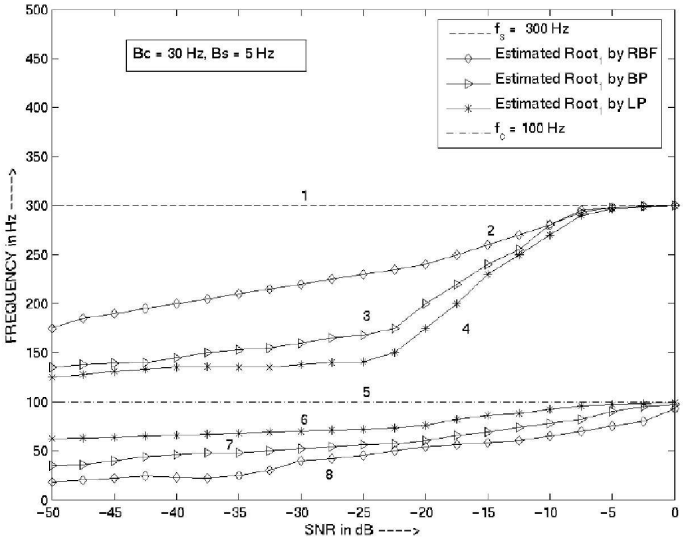


Figure 6: Relationship between estimated frequency roots and SNR for SE,BP and RBF neural network .

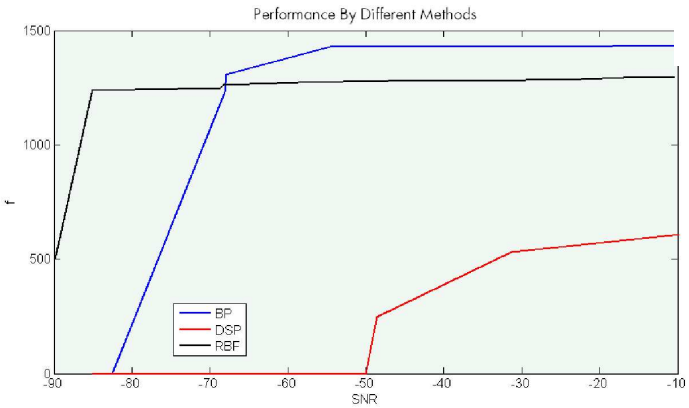


Figure 7: Performance comparison of SE, BP and RBF neural network (Bc = 30 Hz, Bs = 5 Hz).

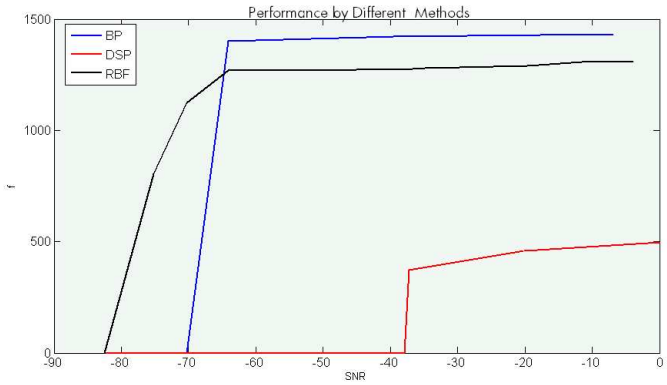


Figure 8: Performance comparison of SE, BP and RBF neural network (Bc = 20 Hz, Bs = 3 Hz).

Table 1: Performance Comparison of SE, BP and RBF MTI.

Algorithms	Signal-to-Sidelobe Ratio, in dB			
	Bc=30Hz Bs=5Hz	Bc=20Hz Bs=3Hz	Bc=10Hz Bs=5Hz	Bc=0Hz Bs=0Hz
SE	-50	-38	-30	-60
BP	-82	-70	-60	-70
RBF	-100	-83	-78	-130

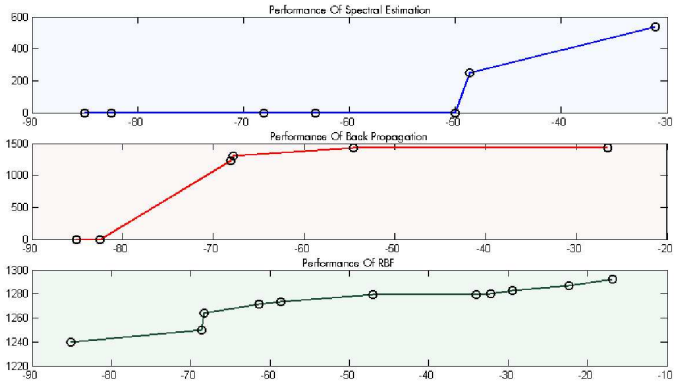


Figure 9: Performance comparison of LP, BP and RBF neural network(Bc = 30 Hz, Bs = 5 Hz).

4 Conclusion

From the simulation results, we concluded that the performance of detection depends on many factors such as the signal to noise ratio, the bandwidths and central frequencies of the noise spectrum, and the signal spectrum. In case of RBF approach we can overcome signal-cancellation problem, as this approach detect signal even at SNR = -100 dB, while SE and BP approaches fails at -50 dB and -82 dB SNR respectively. Together with the high speed computational capability of the neural network, the above analysis and simulations demonstrate the suitability of mix scheme using RBF neural network for moving-target indicator.

References

[1] H. V. Poor, *An Introduction to Signal Detection and Estimation*, Springer-Verlag, New York 1988.

[2] S. A. Kassam, *Signal Detection in Non-Gaussian Noise*, Springer-Verlag, New York, 1988.

[3] E. Wilson, S, Umesh and D. W. Toolts, "Multistage Neural Network Structure For Transient Detection and Feature Extraction", *Proceeding IEEE International Conference On Acoustics, Speech and Signal Processing*, Vol. 1, 1993, PP. 489-492.

[4] R. J. Liou and M. R. Azimi-Sadjadi, "Dim Target Detection Using High Order Correlation Method",

IEEE Transactions on Aerospace and Electronic Systems, Vol.29, No.3, 1995, pp. 841-857.

[5] H. K. Kwan and C. K. Lee, "A Neural Network Approach to Pulse Radar Detection", *IEEE Transactions on Aerospace and Electronic Systems*, Vol.29, No.1, January 1993, pp. 9-21.

[6] R. Deergha and G. Sridhar, "Improving Performance in Pulse Radar Detection Using Neural Networks", *IEEE Transactions on Aerospace and Electronic Systems*, Vol.31, No.3, July 1995, pp. 1193-1198.

[7] M. W. Roth, "Neural Networks for Extraction Of Weak Targets In High Clutter Environments", *IEEE Transactions On Systems, Man, and Cybernetics*, Vol. 19, No. 5, 1989, PP. 1211-1217.

[8] L. Fa-Long and U. Rolf *Applied Neural Networks for Signal Processing*, Cambridge University Press., 1997.

[9] F. L. Luo and J. J. Li, "A Method for Detecting the Moving Target", *Radar*, No. 1, 1991, PP. 12-18.